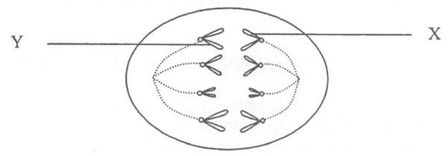
1. 2013Q18

The diagram below shows a dividing cell which is forming an animal's egg cell:

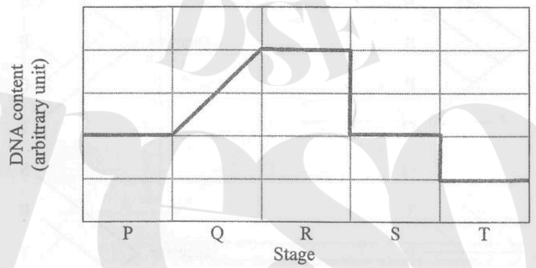


What conclusion about the cell division can be drawn from the diagram?

A. The diagram shows a mitotic cell division.
B. The diagram shows the first meiotic cell division.
C. X and Y are homologous chromosomes.
D. Each daughter cell will have four chromosomes.

2. 2016Q18

Questions 18 to 20 refer to the graph below, which shows the change in the DNA content of a cell undergoing a certain division:



Which of the following can be deduced from the graph?

(1) There are two divisions.
(2) There is one DNA duplication.
(3) DNA content is halved at the end of the whole process.

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

3. 2016Q19

Which of the following stages best represent(s) interphase?

A. P only
B. Q only
C. P and Q only
D. P, Q and R only

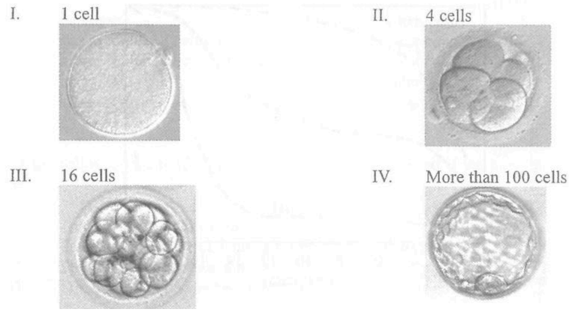
4. 2016Q20

Sister chromatids separate from one another during the transition from

A. P to Q.
B. Q to R.
C. R to S.
D. S to T.

5. 2017Q21

Questions 21 and 22 refer to the following photomicrographs of the same magnification. The photomicrographs show some early stages of embryo development:



Which of the following processes are taking place from stage I to stage IV?

(1) cell division
(2) cell enlargement
(3) cell differentiation

A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3)

6. 2017Q22

How many cell cycles has the cell in stage I gone through to reach stage III?

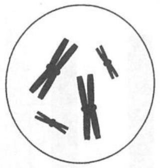
A. 2 cycles
B. 3 cycles
C. 4 cycles
D. 5 cycles

7. 2020Q20

The amount of DNA in cell P immediately before mitosis is x. After division, there are 4 chromosomes in each daughter cell. Which of the following descriptions is correct?

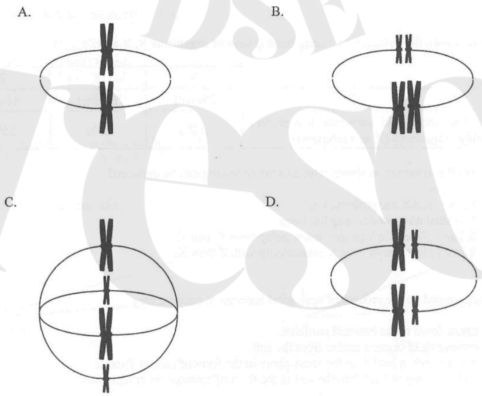
A. The amount of DNA in the daughter cell is 0.5 x.
B. The amount of DNA in each chromosome is 0.25 x.
C. There are 8 chromosomes in the diploid state of cell P.
D. There are 8 chromosomes in cell P immediately before division.

8. 2022Q4
Questions 4 and 5 refer to the diagram below, which shows the appearance of chromosomes in a cell at the early state of meiotic cell division.

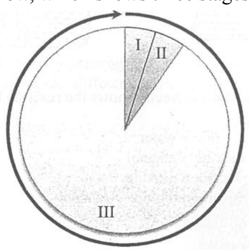


What is the chromosome number in the somatic cells of this organism?
A. 2
B. 4
C. 8
D. 16

9. 2022Q5
Which of the following diagrams correctly shows the arrangement of chromosomes found at a later state of the cell division?



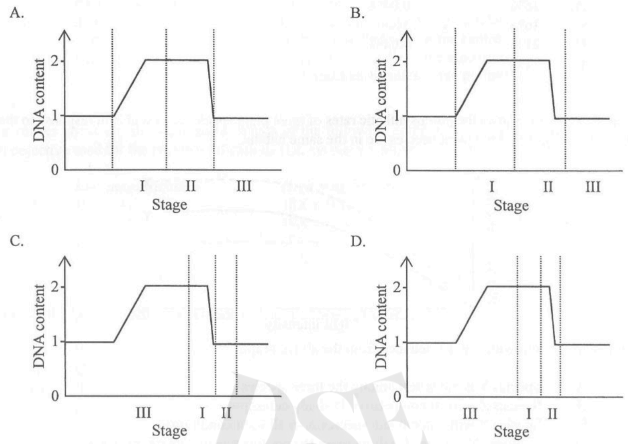
10. 2023Q5
Questions 5 and 6 refer to the diagram below, which shows three stages of a cell cycle:



Which of the following correctly describes the events in the cell cycle?
A. The cytoplasm is halved in stage I.
B. Sister chromatids separate in stage II.
C. The cell synthesises more organelles in stage III.
D. Chromatin condenses to form chromosomes in stage III.

Bio - By topic - Topic 9 Cell cycle and division (2012-2024 version) by **DSERESO**

11. 2023Q6
Which of the following graphs correctly matches the changes in the DNA content in the cell with the three stages of the cell cycle?



12. 2024Q4
Questions 4 and 5 refer to the diagram below, which shows the appearance of chromosomes in a cell at the early stage of meiotic cell division:

If the amount of DNA in this cell is β, what is the amount of DNA in a non-dividing somatic cell of this organism?
A. ¼ β
B. ½ β
C. β
D. 2 β

LQ

1. 2012Q11 - Essay

Mitosis and meiosis are important processes that ensure the continuity of life. Contrast the two processes and state the significance of their differences. (11 marks)

2. 2014Q3

Before the early 20th century, scientists generally held the belief:

"Cell division resulted in the loss of genetic material so that each cell in a multicellular organism would contain only the genetic material specific to its particular cell type."

In 1902, Hans Spemann performed one of the earliest experiments on animal cloning. He used a fine hair to separate the cells of a two-celled amphibian embryo, and found that each cell was able to develop into a complete organism.

a) Why did Spemann's experiment disprove the early belief about cell division? (1 mark)

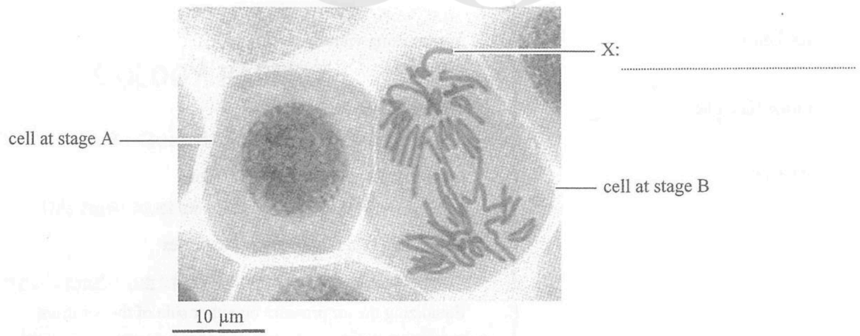
b) Elaborate on how the above example can be used to demonstrate the two aspects of the nature of science listed in the table below.

Nature of Science	Elaboration
Scientific knowledge is tentative and subject to change.	
Interpretation of observations is guided by our prior understanding of other theories and concepts.	

c) Using the current understanding about cell division, explain how genetic material is preserved in mitosis. (3 marks)

3. 2015Q2a, c

The photomicrograph below shows the appearance of genetic materials at two different stages of the cell cycle:



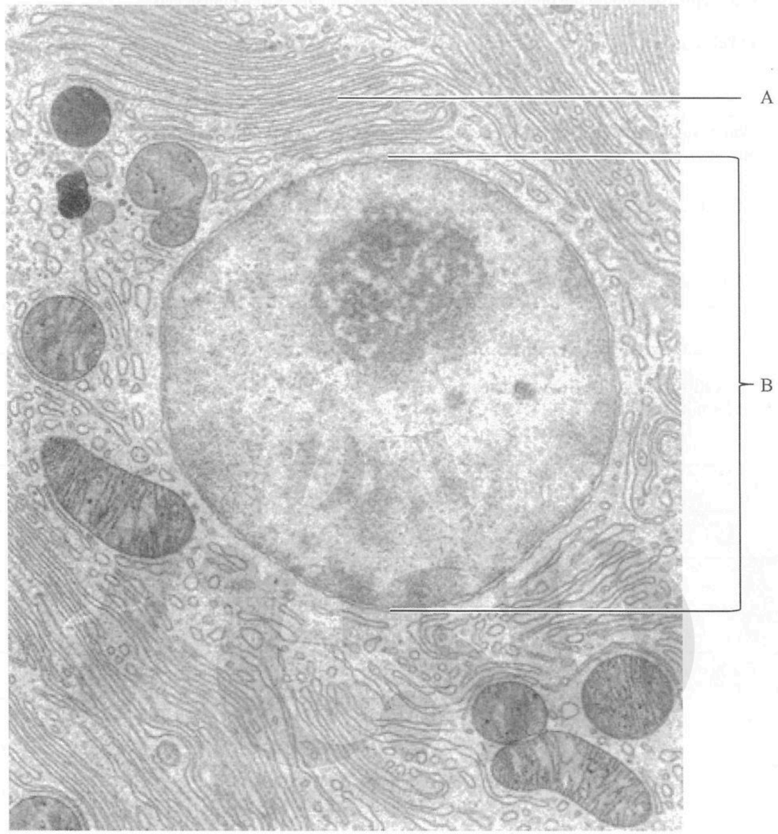
a) Label structure X shown in the photomicrograph. (1 mark)

c) In the space provided below, state the cause for the different outcomes of mitosis and meiosis. (2 marks)

	Outcome		***Cause***
	Mitosis	***Meiosis***	
Number of daughter cells	2	4	
DNA content in daughter cells	2N	1N	

Bio - By topic - Topic 9 Cell cycle and division (2012-2024 version) by **DSERESO**

4. 2017Q4b

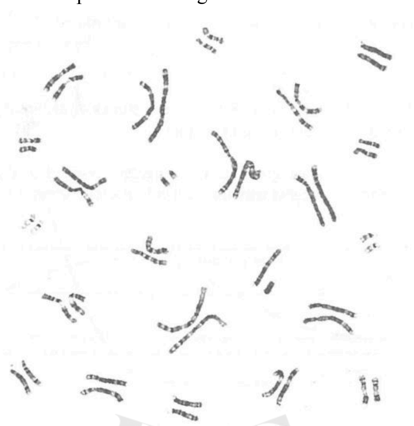


The electron micrograph shows some structure of a human cell.

b) Which stage of the cell cycle is shown in this photomicrograph? Give a reason to support your answer. (2 marks)

5. 2018Q5a, b

The photomicrograph below shows the paired homologous chromosomes of a normal boy for karyotyping:



a) Circle the sex chromosomes on the above photomicrograph. (1 mark)

b) State the type of cells, somatic cells or gametes, from which the karyotype was obtained. Explain your answer. (2 marks)

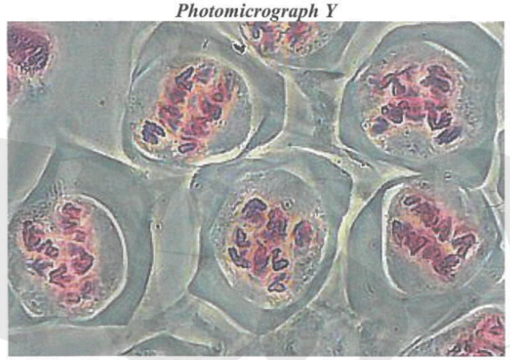
Bio - By topic - Topic 9 Cell cycle and division (2012-2024 version) by **DSERESO**

6. 2019Q3b, c

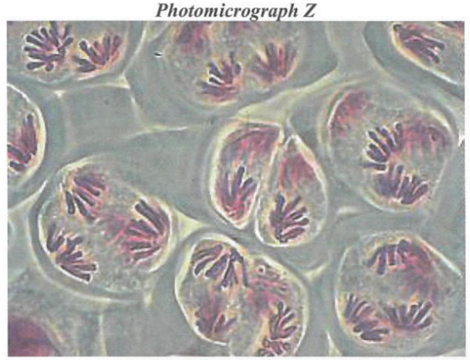
The photomicrographs below show some stage of meiosis taking place in a flower:

Photomicrograph X

Photomicrograph Y



Photomicrograph Z



b) i) Name event W shown in Photomicrograph X. (1 mark)

ii) Briefly describe what happens in event W. What is the importance of event W? (2 marks)

c) i) Which photomicrograph, Y or Z, shows the first meiotic division? Give a piece of evidence to support your answer. (2 marks)

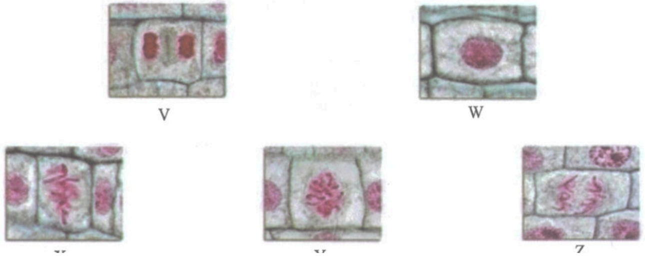
ii) What is the purpose of the first and second meiotic divisions respectively? (2 marks)

7. 2021Q5b, c

A student prepared cells of an onion root tip for observing cell division under a light microscope.

b) Suggest ***one*** necessary step to make the chromosomes observable under a light microscope. (1 mark)

c) Some events of the cell division are randomly shown in the following photomicrographs:



i) Starting with photomicrograph W, arrange the photomicrographs in the correct order to show the sequence of events in cell division. (1 mark)

W → _______ → _______ → _______ → _______

ii) A normal onion root cell has 16 chromosomes. Complete the following table to show the number of chromosomes and chromatids in photomicrographs Y and Z. (2 marks)

Photomicrograph	Number of chromosomes	Number of chromatids
Y		
Z		

Bio - By topic - Topic 9 Cell cycle and division (2012-2024 version) by **DSERESO**

Ans

2013Q18: D (32%)	2016Q18: D (67%)	2016Q19: C (34%)	2016Q20: D (59%)
2017Q21: B (53%)	2017Q22: C (63%)	2020Q20: A (34%)	2022Q4: B (66%)
2022Q5: B (44%)	2023Q5: C (48%)	2023Q6: D (42%)	2024Q4: B (41%)

2012Q11

Differences	***Significance***	
• Pairing of homologous chromosomes along the equatorial plane in first division of meiosis but no such process in mitosis (1) • The pairs of homologous chromosomes segregate into the daughter nuclei during the first meiotic cell division (1)	• Such that the daughter cells formed contain the whole set of chromosome / one member of each homologous pairs (1) after meiosis • Random segregation of homologous chromosomes results in variations between gametes formed (1) • Crossing over may occur, the exchange of genetic materials between chromatids gives rise to new genetic combinations (1)	(5)
• Mitosis involves one division only but meiosis involves two divisions (1)	• The daughter cells resulted from mitosis are genetically identical to the parent cell (1), which is important for growth of the organisms (1) / asexual reproduction • The daughter cells / gametes formed in meiosis contain half / haploid the genetic content of the parent cell (1) such that the amount of genetic content can be restored after fertilisation (1)	(5)
D = (3)	S = (max. 5)	max. 8

Communication (C) max. 3

11 marks

2014Q3

3 (a) • since each of the separated cells was able to develop into a complete organism, this implies a whole set of genetic material is present in each cell / there was no reduction in the quantity of genetic material during the first division (1) (1)

(b)

Science knowledge is tentative and subject to change.	Spemann's results disproved the general belief, showing that scientific knowledge will change when there is new evidence evolved (1)	(1)
Interpretation of observation is guided by our prior understanding of other theories and concepts	Scientist observed that cell division and believed that all the materials inside will be divided too without knowing that genetic materials will be duplicated (1) / Spemann thought that all genetic materials are required to form a whole organism; therefore, he interpreted that each cell contained a complete set of genetic material (1)	(1)

(c) • DNA / genetic material is duplicated right before cell division (1) (1)

• the duplicated chromosomes will line up at the middle part of the cell for separation (1) (1)

• each member of the duplicated chromosomes will then separate and eventually divide equally into each daughter cell (1) (1)

6 marks

2015Q2a, c

2 (a) • chromosomes / chromatid (1) (1)

(c)

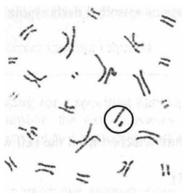
	Outcome		Cause	
	Mitosis	Meiosis		
Number of daughter cells	2	4	mitosis involves one division while meiosis involves two divisions (1)	(2)
DNA content in daughter cells	2N	1N	homologous chromosomes pair up and separate into each daughter cell in meiosis but not in mitosis (1)	

2017Q4b

4 (b) • interphase (1)
• because DNA is in the format of chromatin / the chromosomes are invisible (1) (1+1)

2018Q5a, b

5 (a) •



(1)

(b) • somatic cells (1)
• because there are 2 sets of chromosomes / 46 chromosomes / 23 pairs of chromosomes (1) (1+1)

2019Q3b, c

3 (b) (i) • crossing over (1) (1)

(ii) • exchange of genetic materials between non-sister chromatids of the homologous pair of chromosomes (1)
• This is an important source of genetic variation 91) (2)

(c) (i) • Photomicrograph Y belongs to the first meiotic cell division (1)
• because pairing or separation of homologous chromosomes is shown (1), which is the characteristic feature in first meiotic cell division (2)

(ii) • first meiotic cell division separates the two sets of homologous chromosomes (1)
• while second meiotic cell division separates the sister chromatids (1) (2)

8 marks

2021Q5b, c

5 (b) • use a stain to stain the chromosomes / staining / any names stain (1) (1)

(c) (i) W → Y → X → Z → V (1 or 0) (1)

(ii)

Stage	Number of chromosomes	Number of chromatids	
Y	16	32	(1)
Z	32	0	(1)

6 marks